

GENERAL SPECIFICATION G2.5

Condition Surveys

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G2.5 CONDITION SURVEYS

G2.5.1 Government Survey Datum

- (a) All levels shown on drawings shall be taken with reference to the Singapore Height Datum of 0.000m for measurement of vertical distance.

G2.5.2 Pre-Construction Surveys

- (a) The Contractor shall, prior to commencement of the Works, engage qualified firm/personnel to carry out condition survey for all existing properties, buildings and structures, LTA Structures and Road Elements, and other adjacent structures within the influence zone from the contract boundary as stipulated in the latest BCA guidelines and as specified.
- (b) As part of the scope for the condition surveys, the qualified firm/personnel shall jointly carry out the following scope of the condition survey work:

- (i) Determine the owners of all the properties adjacent to the Works as directed by the S.O. and make arrangements with the owner, or his authorised representative to carry out a condition survey of his property. In the event that the Contractor is unable to gain entry to the properties in the zone of preconstruction survey, the contractor shall notified the S.O. and comply with the BCA guidelines to:

- 1) Survey the exterior face of the property;
 - 2) Keep records of attempts to contact relevant owners for permission to conduct preconstruction survey (e.g. records of registered mail); and
 - 3) Keep record of refusal by owner to allow access to conduct survey.

Notwithstanding the above, the Contractor shall remain responsible for any damage arising from the construction work irrespective of whether the survey was carried out for the damaged premises.

- (ii) Carry out a condition survey of each of the property to the satisfaction of the S.O.. The condition survey shall include but not be limited to the following:
 - 1) A visual and photographic survey of both the external and internal structures of the property including all floor & basement levels; and
 - 2) A report on the extent of the survey carried out for each property. The report shall contain the following information for each property:
 - Property name, if any and type;
 - Property location;
 - Name of Owner, and his authorised representative; and
 - Date and details of the survey. This shall include but not be limited to the following:
 - Areas of the property surveyed and areas where it was not possible to survey, together with the reasons why;

- Areas of the property have been jointly surveyed with the owner or his authorised representative, together with any areas surveyed independently;
 - To state in the report the area or a section of the property that has been surveyed and no sign of damage or deterioration has been found;
 - To illustrate the damage or deterioration observed using plans and sketches showing the location, nature and extent. This will include but not necessarily be limited to the length, width and depth of existing cracks; and
 - Dated colour photographs on 35mm film of every elevation, areas of sensitive details, and areas where there is already visual signs of damage or deterioration. Each photograph in the report shall be clearly titled and shall have a brief description to define its exact location.
- (c) The Contractor shall identify and record all existing defects and damages of the properties, buildings and structures including but not limited to cracks, settlement, concrete spalling, seepage, etc. The Contractor shall carry out his own assessment based on his detailed construction methodology and the predicted ground movements and other influence factors to determine the details and areas which are likely to be affected by the Works in his condition surveys. Such areas shall include but not limited to all structural and non-structural components of each property, structure and building that are deemed necessary.
- (d) Every attempt shall be made to survey all premises likely to be affected by the construction work and the Contractor shall remain responsible for any damages arising from the construction work irrespective of whether the survey was carried out at the damaged premises.
- (e) The Contractor shall submit 2 softcopies (in CD) and 2 hardcopies of the survey reports (endorsed by the Contractor and the qualified firm/personnel) to the Board/ to the S.O. within three(s) months from the date of site possession and prior to commencement of the Works.
- (f) The Contractor shall forward the owner a copy of the pre-construction survey report on his/her property before any commencement of work. The Contractor shall ensure that the owner acknowledge the receipt of this report.
- (g) At the stage of applying for the Permit to commence piling/building works, the Contractor shall give an undertaking to the Building and Construction Authority (BCA) as per format required by BCA to confirm that a copy of the pre-construction survey report shall be given to each of the owners of the neighbouring properties before work commences. A copy of the pre-construction survey report is to be kept at site by the Contractor.
- (h) The acceptance of the pre-construction survey report is subjected to the satisfaction of the S.O. and the survey/report shall be re-taken/re-submitted at the Contractor's own cost in the event that the report is not acceptable to the S.O..

- (i) An item is provided in the Bills of Quantities for the pre-construction survey reports to be submitted. Upon completion of the work, the contractor shall carry out a post-work condition survey of the properties adjacent to the works together with any additional properties affected by the construction works.

G2.5.3 As-Built Topographical Survey for Works Outside of the Board's Facility

- (a) General
 - (i) Upon completion of works external to the Board's Facility boundary, the Contractor shall engage a land surveying firm registered with the Land Surveyors Board, Singapore to undertake the as-built survey works. The surveyor shall be a competent Registered Surveyor holding a valid practicing certificate issued by the Land Surveyors Board, Singapore.
 - (ii) The Tender sum shall be deemed to have included but not limited to all cost such as finding and providing labour, instruments, materials, insurance, plant, transport, tools, plans and report printing and all else necessary to carry out the survey.
- (b) Scope of Works
 - (i) To carry out topographical survey and preparation of plans for full extent of the areas under the scope of this project.
 - (ii) The Surveyor shall carry out the following works and all other works incidental for the proper completion of the survey work:
 - 1) Survey and produce topographical plans based on the SVY 21 Co-ordinate System to a scale of 1:200 of the areas within and at least six (6) metres beyond the project site boundary;
 - 2) The topographical survey drawings in electronic format must be compatible to MicroStation (3D) format (latest version). The drawing files with survey points plotted in 3-dimensional (3D) scale should also follow the level and colour schemes as specified in Softcopy Submission;
 - 3) To produce a location plan to the scale of 1:5000;
 - 4) To produce cross-section drawings to a scale of 1:500 at every location where proposed river cross sections are provided in the Drawings;
 - 5) All survey shall refer to official government benchmarks, which shall be indicated in the plans;
 - 6) The surveyor shall also incorporate the proposed site boundary co-ordinates and other reference co-ordinates onto his plan; and
 - 7) The surveyor shall include in his pricing the purchase of any services and cadastral plans. The services lines and cadastral boundaries shall be mapped onto separate surveyed drawings.
 - (iii) Reference pegs with markings for identification on site shall be established for the base line and all intersection points, including all those in (d) above. These pegs shall be based on permanent and prominent structures or any other locations which will not be destroyed for a reasonable period of time.

- (iv) Where no convenient feature is available, the Surveyor shall establish his own reference pegs. The location and nature of all reference pegs and their relationship with the setting out pegs shall be clearly shown in the final drawing(s).
- (v) The base line shall consist of straight lines only, with their bearings and lengths shown on site plan. The base lines should be related to the boundary stones.
- (c) Deliverables
 - (i) The surveyor shall supply the following materials during and on completion of the works.
 - 1) Softcopy Submission – Two copies of CD ROM shall be submitted. The survey plans and verified related information with all its contents shall be in soft copy and suitable for use with MicroStation (latest version). Each CD-ROM shall have one copy of each of the following:
 - Topographical survey plans to a scale of 1: 250;
 - Key Plan to a scale of 1:5000;
 - All survey points shall be plotted in the x-, y- and z- planes;
 - Cross-sections to a scale of 1:500 of the surveyed area at every location where proposed river cross sections are provided in the Drawings;
 - The softcopy plan shall contain all layers and textual information of the surveyed items as indicated in the hardcopy;
 - Contour lines of 1m intervals (or appropriate intervals depending on the terrain of the site);
 - Separate softcopy of survey plan showing the entire surveyed area without title block;
 - Underground Services Plans from the agencies as specified;
 - Road Line Plans from LTA; and
 - Cadastral Plans from SLA.
- (d) Softcopy submission requirements are as follows:
 - (i) The media for data submission shall be speed 8x to 12x compatible CD-R (Compact Disk Recordable);
 - (ii) The survey plans and verified related information with all its contents shall be submitted in soft copy Microstation format, latest version (dgn. or .dxf);
 - (iii) The softcopy plan shall contain all layers and textual information of the surveyed items as indicated in the hardcopy plan;
 - (iv) The softcopy plan shall contain separate layers for each type of surveyed items;
 - (v) The drawing files shall follow the level scheme, colour assignment as specified in Singapore Standard CP 83, Code of Practice for Construction Computer-Aided Design (CAD) Parts 1-5 as well as the requirements set out below; and
 - (vi) Layer Naming Convention as per P2 shall be followed.

- (e) Hardcopy Submission – Hardcopy submissions as listed below, showing all the required information in the Specifications shall be duly endorsed by a Registered Surveyor:
 - (i) Two (2) sets of topographical survey plans to a scale of 1: 250 printed on A1 size paper with spot levels indicated and Location Plan included;
 - (ii) One (1) set of lot boundary plans indicating boundary stones, Lot and TS/Mukim boundaries and numbers printed on A1 size paper;
 - (iii) One (1) set of services plans to a scale of 1:200 incorporating all underground services, printed on A1 size paper;
 - (iv) Longitudinal sections for the both sides of the river, showing its existing levels at 10m intervals for the drains found within the surveyed area. For these longitudinal sections, the horizontal scale of 1:250 and vertical scale of 1:50 is to be used; and
 - (v) Original services plan from the agencies as specified, purchased for the survey works.
- (f) Programme
 - (i) The Contractor's surveyor shall carry out the survey of the site and submit final endorsed copies of plans before Substantial Completion of the Works.
- (g) Specifications
 - (i) Generally, the finished final survey drawings of the demarcated area shall show the following:
 - 1) Demarcation and orientation of the existing site boundaries (bearings and distances between boundary pegs to be indicated);
 - 2) Spot levels at grid intervals not exceeding 5 metres;
 - 3) Contour lines at 1 metre intervals;
 - 4) Location, type, route and invert levels (if applicable) of all existing and exposed services like electric cables, water pipes, gas pipes, sewer pipes, etc.;
 - 5) Location, common name of trees, scientific name of trees, height, girth and canopy size drawn to scale of all existing trees and plants. The Contractor's Landscape Architect and Tree Arborist shall verify the correctness of all tree names. The trunks of individual trees with girth exceeding 500mm measured 1.0m above ground level shall be numbered with markers approved by NParks and indicated with number in the plan;
 - 6) Location, type, and coverage of all bioengineering techniques used in the river;
 - 7) Configuration and platform levels of all existing buildings, carparks (with parking lots indicated), driveways and other special features;
 - 8) Location of all boundary stones, Lot and TS/Mukim boundaries and numbers;

- 9) Location, type, size and top of cover and base slab level of all trenches, manholes, junction and inspection chambers, drains, culverts, sumps, footpaths and roads etc. within the survey area;
 - 10) The surveyor shall indicate the sewer top and inlet levels on the plan. The surveyor shall be responsible to coordinate with the relevant departments for assistance to open the manhole cover if required;
 - 11) Diameter, invert levels and horizontal positions of incoming pipes, backdrops, tumbling bays connecting to existing manholes;
 - 12) Other landmarks which can be clearly indicated on the survey plan;
 - 13) Location of reference benchmarks complete with co-ordinates and reduced levels. Relevant northing and easting co-ordinates shall also be plotted accordingly;
 - 14) Calculation of the site area within the boundaries and the actual area surveyed;
 - 15) Cross-sections at every location where proposed river cross sections are provided in the Drawings. The locations of the Cross sections shall be indicated on the plans. The cross sections shall include all prominent features captured on the plan; and
 - 16) Services plans including, but not limited to, road interpretation plan, MRT/LRT reserve plan, plant maps, maps indicating existing gas main and maps area indicating electricity cables in the surveyed area are to be obtained from the following agencies:
 - Land Transport Authority;
 - Singapore Telecommunications Ltd;
 - Power Gas;
 - Power Grid;
 - Singapore Land Authority;
 - Singapore's Mass Rapid Transit; and
 - Starhub.
- (ii) Drainage Interpretation Plans will be provided to the Surveyor. The Drainage Reserve Lines shall be reflected clearly as specified but not limited to condition 5.2 below with legend for the different services; and
- (iii) Sewer Interpretation Plans will be provided to the Surveyor for information to assist in locating the existing manholes within the site.
- (h) The Surveyor is required to locate accurately and mark on separate layer/plan the exact location of the underground sewer lines, water line and all other relevant services within the demarcated area.
- (i) The site area and the actual area surveyed must be clearly stated in the final survey plans, and the actual number of boundary pegs established must be clearly marked both on the site and in the final survey plans for verification purposes.

- (j) The co-ordinate system for the digitized planimetric plans shall follow the SVY21 Co-ordinate System. The elevation information shall conform to the Government Precise Level.
- (k) All survey shall refer to official government benchmarks, which shall be indicated in the plans.
- (l) The surveyor shall also incorporate the proposed site boundary co-ordinates and other reference co-ordinates onto his plan.
- (m) Hydrographic Survey
 - (i) Hydrographical survey shall be carried out where necessary to measure the profile of bed surface within surveyed area if any and the profile shall be drawn on cross-sections.
 - (ii) The Surveyor shall use a suitable electronic sounding equipment according to the following specifications:
 - 1) Production of Sounding Plan shall have sounding depth spaced at 4mm apart or smaller by interpolation between fixed point along sounding line. The number of cross-sections required shall also be produced; and
 - 2) Tidal corrections shall be applied to soundings at interval sufficiently close to meet the survey accuracy of (+) or (-) 0.1 metre.
- (n) Reference Pegs
 - (i) Reference pegs with markings for identification on site shall be established for the base line and all intersection points, including government benchmarks. These pegs shall be based on permanent and prominent structures or any other locations which will not be destroyed for a reasonable period of time.
 - (ii) Where no convenient feature is available, the Surveyor shall establish his own reference pegs. The location and nature of all reference pegs and their relationship with the setting out pegs shall be clearly shown in the final drawing(s).
 - (iii) The Surveyor shall also locate existing boundary stones/ pegs on site and indicate on the plans.
 - (iv) The base line shall consist of straight lines only, with their bearings and lengths shown on site plan. The base lines should be related to the boundary stones.
- (o) Mapping
 - (i) Unit of Measurement

Metric units shall be used throughout.
 - (ii) Contours

Contours, where required shall be shown at vertical interval of 1.0m at 1:1000 scale or equivalent scale in 1:200 or 1:500 appropriate to capture the whole site boundary.
 - (iii) Ground Controls (Transverse and Level)

Ground control of accuracy 1:8000 shall be carefully established so as to ensure the accuracy of the field survey.

(iv) Grid and Height datum

- 1) The plans shall be produced on the same co-ordinate framework currently adopted by Singapore Land Authority, i.e. SVY 21 Co-ordinate System;
- 2) Grids shall be marked with cross marks at 50m centres referenced at 100 metres interval around and at edges of detail; and
- 3) The elevation information shall be in conformity with Government Precise Value.

(v) Standard Symbols to be used

The Standard Symbols used in the existing 1:2500 maps of the 1969/70 series, or any latest updates, shall be adopted.

(vi) Spot Levels

- 1) Levels shall, generally, be taken and shown in the plan at 5m grid intervals within the site boundary lines. Notwithstanding the above, where embankments and slopes are encountered, levels of the top and bottom slope are to be taken and indicated on the plan; and
- 2) Where the grid lines cross the roads, drains, etc., within and beyond the boundary lines, the top and invert levels shall also be shown.

(p) Drawing Standard

- (i) Drawing shall be produced on standard sheets of 594mm x 840mm (A1 size) unless otherwise directed.

(q) Details to be clearly informed and shown on Survey Plans (where applicable):

(i) Buildings

- 1) The plinth line of all permanent buildings;
- 2) Open sides of buildings – by picked lines;
- 3) Huts irrespective of floor areas;
- 4) Garden sheds;
- 5) Glass sheds;
- 6) Glass houses;
- 7) Covered agricultural plot;
- 8) Ruins or partially demolished buildings or foundations – by the walls and masonry visible at the time of survey;
- 9) Numbers of all domestic premises or plots; and
- 10) Property boundary lines, Structure ID/names, owner of property.

(ii) Drainage Reserve

- 1) Spot Levels of the cope of existing canals shall be clearly shown at 10m intervals;
- 2) The edge of existing canals shall be clearly shown;

- 3) Drainage reserves including the edge of the drain / canal wall, width, top level, invert level of the drain/ canal, the top level, invert levels and cross-sections at different profiles of the drainage reserves;
 - 4) Depth and direction of flow in all drains and canals and the width and invert levels of these features at 10m longitudinal spacing within the survey area;
 - 5) Location, type, route and size of all existing services crossing the drain/canal shall be clearly shown on Survey Plans; and
 - 6) The Drainage Reserve line shall be clearly shown on the Survey Plans.
- (iii) HDB developments
- 1) The edge of the HDB developments shall be clearly shown on the plan; and
 - 2) Spot Levels of the existing HDB block shall be clearly shown.
- (iv) Location and level of the top and bottom of kerbs and the crown of all roads, run-ways and driveways at 10m intervals along the road or drive. The survey shall indicate the minimum under-clearance for all elevated structures such as footbridges, sign gantries and the following items etc.
- (v) Roads, Tracks and Footpath
- 1) Kerb lines or edge of surfacing to carriage way;
 - 2) Track and bridge paths;
 - 3) Railway tracks;
 - 4) Rides and drives;
 - 5) Culverts (specify size and type);
 - 6) Footways; and
 - 7) Railings.
- (vi) Road Reserve
- 1) Road reserves including the carriageway, drain, footpaths, lampposts, bollard, bus-shelters, bus bay, traffic signs, OG boxes, covered linkway and any other structures etc. Cross sections must be shown at different profiles of the road reserve and road name to be indicated; and
 - 2) Detailed dimension survey both in plan and elevation, on the traffic sign, covered linkway and any structures etc. shall be reflected in the drawings.
- (vii) MRT Reserve
- 1) MRT reserves including the tunnel/viaduct centre-line, first reserve, second reserve, etc.
- (viii) Other Details/ Features
- 1) Valve chambers;
 - 2) Transformers (boundary fences only);
 - 3) Electricity substations (boundary fences only);

- 4) Electricity poles;
- 5) OG Boxes;
- 6) Water Keyboxes;
- 7) Electric Boxes;
- 8) Telegraph lines;
- 9) Telegraph poles (with number);
- 10) Earth drains;
- 11) Concrete drain (specify size and type);
- 12) Fences / Railings (specify height);
- 13) Gates;
- 14) Lamp posts (with number) etc;
- 15) Level crossing;
- 16) Signals;
- 17) Warning signs;
- 18) Hydrants, public telephone posts, post boxes;
- 19) Water and gas valves;
- 20) Slab over drains parapet walls;
- 21) Underground culverts (with description, size and directions of flow);
- 22) Retaining walls (specify height and type);
- 23) Longitudinal cross section, elevation and height clearance of all bridges including bridges over canal;
- 24) Ponds and other bodies of water/streams and any other permanent features;
- 25) Slopes of embankments or cutting with height greater than 1.0m but of too steep a gradient to be indicated by contours are to be shown by conventional markings and the top and bottom of slopes are to be shown as dotted lines;
- 26) Slopes conventions shall be drawn as near as possible to indicate the actual slope of the slopes faces i.e. all beams and terraces etc., are to be detailed;
- 27) Landscape structures, temporary and permanent structures, shelters, boundary fencing, building and retaining wall, children playground etc; and
- 28) And any other features on site.

- (r) Field Survey
 - (i) All control stations or points of intersection shall be pegged with either iron pipes or rods set in concrete. Their exact locations and reduced levels should be shown in the setting out plan and other detail drawing with bearings and distances from its adjacent point and from its recovery pegs. From this traverse, he shall establish the intersection points including reference marks.
 - (ii) At least three (3) permanent bench marks should be set up within each area of survey, in addition to all the temporary bench marks provided by the Surveyor. The reference of orientation for each bench mark shall be at least three (3) prominent reference objects of no less than 100m apart. All benchmarks shall be included in topographical survey.
- (s) Security and Authorisation to Enter Premises
 - (i) The Contractor or his Surveyor shall be responsible for obtaining the necessary clearance to enter any areas to be surveyed.
- (t) Security
 - (i) In the event of any worker found tampering, damaging or removing public or private property or materials without approval, that person is liable to be arrested and appropriate action taken.
- (u) Plan Certification
 - (i) The qualified Registered Surveyor/Licensed Surveyor shall endorse on every final survey plans submitted.
- (v) Omission
 - (i) The Surveyor shall note any omission of the work should be consulted and approval obtained from the S.O. before commencement of the work.
- (w) Materials and Workmanship
 - (i) In general, all materials and workmanship used in the survey works and in the production of the finished maps shall be the best of their respective kinds and shall comply with either the British Standard Specification or Singapore Standards, whichever is appropriate.
 - (ii) All materials prepared and/or obtained during the course of the survey works shall remain the property of the Board and may not be reproduced without the prior written permission of the Board.

END OF SECTION